PRAKTIKUM BASIS DATA

MODUL 8

“Scripting SQL”

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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UNIVERSITAS TELKOM

2026

JURNAL

Soal

MODUL 8

JURNAL

Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban:

1. department :

```
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM barang;
SELECT * FROM persediaan;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.002 seconds

DEPARTM...	DEPARTM...	MANAGER...	LOCATIO...

Employee :

```
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM barang;
SELECT * FROM persediaan;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.001 seconds

EMPLOYE...	LAST_NAME	EMAIL	SALARY	COMMISS...	HIRE_DATE	DEPARTM...

Supplier :

```
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM barang;
SELECT * FROM persediaan;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.003 seconds

SUPPLIER...	SUPPLIER...

Barang:

```
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM barang;
SELECT * FROM persediaan;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.001 seconds

BARANG_ID	BARANG_...	SUPPLIER...

Persediaan:

```
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM barang;
SELECT * FROM persediaan;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.002 seconds

PERSEDIA...	BARANG_ID	JUMLAH

2.

```
BEGIN
    DBMS_STATS.GATHER_SCHEMA_STATS(ownname => USER);
END;
/
```

Script Output

Task completed in 11.742 seconds

```
PL/SQL procedure successfully completed.
```

3.

Department :

```
INSERT INTO department VALUES (60, 'Logistik', 106, 1006);
INSERT INTO department VALUES (70, 'Produksi', 107, 1007);
INSERT INTO department VALUES (80, 'Support', 108, 1008);
INSERT INTO department VALUES (90, 'R&D', 109, 1009);
INSERT INTO department VALUES (100, 'Legal', 110, 1010);

SELECT * FROM department;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.025 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1001
2	20	IT	102	1002
3	30	Finance	103	1003
4	40	Marketing	104	1004
5	50	Sales	105	1005
6	60	Logistik	106	1006
7	70	Produksi	107	1007
8	80	Support	108	1008
9	90	RD	109	1009
10	100	Legal	110	1010

Employee :

```
INSERT INTO employee VALUES (6, 'Fajar', 'fajar@mail.com', 4500, 0.1, DATE '2022-04-25', 40);
INSERT INTO employee VALUES (7, 'Gina', 'gina@mail.com', 4800, 0.2, DATE '2023-05-10', 50);
INSERT INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 5200, 0.1, DATE '2023-06-15', 60);
INSERT INTO employee VALUES (9, 'Indah', 'indah@mail.com', 5300, 0.2, DATE '2023-07-20', 70);
INSERT INTO employee VALUES (10, 'Joko', 'joko@mail.com', 6200, 0.15, DATE '2023-08-01', 80);
SELECT * FROM employee;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.003 seconds

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSIO...	HIRE_DATE	DEPARTMENT_ID
1	1	Andi	andi@mail.com	5000	0,1	01-01-2020	80
2	2	Budi	budi@mail.com	6000	0,2	01-02-2020	80
3	3	Citra	citra@mail.com	5500	0,15	10-01-2021	10
4	4	Dedi	dedi@mail.com	7000	0,1	15-02-2021	20
5	5	Eka	eka@mail.com	8000	0,05	20-03-2022	30
6	6	Fajar	fajar@mail.com	4500	0,1	25-04-2022	40
7	7	Gina	gina@mail.com	4800	0,2	10-05-2023	50
8	8	Hadi	hadi@mail.com	5200	0,1	15-06-2023	60
9	9	Indah	indah@mail.com	5300	0,2	20-07-2023	70
10	10	Joko	joko@mail.com	6200	0,15	01-08-2023	80

Supplier:

```sql
INSERT INTO supplier VALUES (7, 'Supplier G');
INSERT INTO supplier VALUES (8, 'Supplier H');
INSERT INTO supplier VALUES (9, 'Supplier I');
INSERT INTO supplier VALUES (10, 'Supplier J');
SELECT * FROM supplier;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 10 in 0.003 seconds

	SUPPLIER_ID	SUPPLIER_NAME
1	1	Supplier A
2	2	Supplier B
3	3	Supplier C
4	4	Supplier D
5	5	Supplier E
6	6	Supplier F
7	7	Supplier G
8	8	Supplier H
9	9	Supplier I
10	10	Supplier J

Barang:

```sql
INSERT INTO barang VALUES (7, 'Router', 7);
INSERT INTO barang VALUES (8, 'Switch', 8);
INSERT INTO barang VALUES (9, 'Flashdisk', 9);
INSERT INTO barang VALUES (10, 'Harddisk', 10);
SELECT * FROM barang;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 10 in 0.001 seconds

	BARANG_ID	BARANG_NAME	SUPPLIER_ID
1	1	Laptop	1
2	2	Mouse	2
3	3	Keyboard	3
4	4	Monitor	4
5	5	Printer	5
6	6	Scanner	6
7	7	Router	7
8	8	Switch	8
9	9	Flashdisk	9
10	10	Harddisk	10

Persediaan:

```
INSERT INTO persediaan VALUES (6, 6, 35);
INSERT INTO persediaan VALUES (7, 7, 40);
INSERT INTO persediaan VALUES (8, 8, 50);
INSERT INTO persediaan VALUES (9, 9, 60);
INSERT INTO persediaan VALUES (10, 10, 70);
SELECT * FROM persediaan;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.002 seconds

	PERSEDIAAN_ID	BARANG_ID	JUMLAH
1	1	1	10
2	2	2	20
3	3	3	30
4	4	4	15
5	5	5	25
6	6	6	35
7	7	7	40
8	8	8	50
9	9	9	60
10	10	10	70

4. analyze

```
BEGIN
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'DEPARTMENT');
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEE');
END;
/
```

Script Output

Task completed in 0.16 seconds

```
PL/SQL procedure successfully completed.


PL/SQL procedure successfully completed.
```

5. view

```
CREATE VIEW empuv80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

Script Output | Task completed in 0.064 seconds

```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

View EMPUV80 created.
```

6. view emvu80

```
WHERE department_id = 80;
SELECT * FROM empuv80;
```

Script Output | Query Result | SQL | All Rows Fetched: 3 in 0.004 seconds

	ID_NUMBER	NAME	SALARY	DEPARTMENT_ID
1	1	Andi	5000	80
2	2	Budi	6000	80
3	10	Joko	6200	80

7.

```
CREATE INDEX idx_dept_id
ON employee(department_id);
```

Script Output | Query Result | Task completed in 0.053 seconds

```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

View EMPUV80 created.

Index IDX_DEPT_ID created.
```

```
DROP INDEX idx_dept_id;
```

Script Output x | Query Result x

Task completed in 0.171 seconds

```
PL/SQL procedure successfully completed.


PL/SQL procedure successfully completed.


View EMPUV80 created.


Index IDX_DEPT_ID created.


Index IDX_DEPT_ID dropped.
```